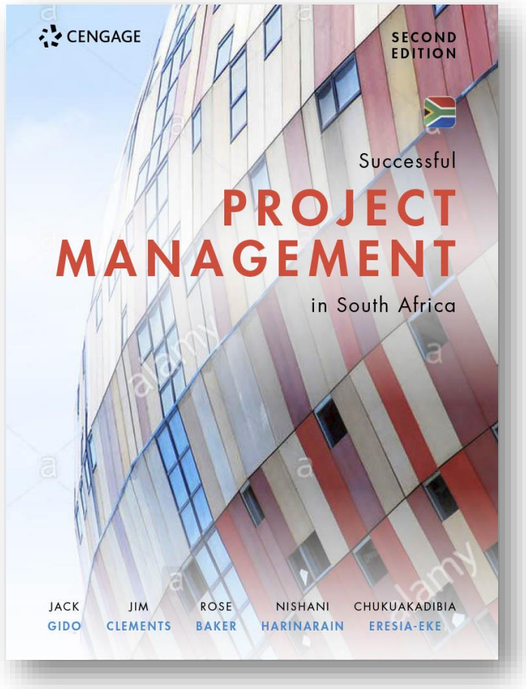
IIE BACHELOR OF COMPUTER AND INFORMATION SCIENCES IN APPLICATION DEVELOPMENT (BCA2)



INFORMATION SYSTEMS 2B

INSY6212w

Felicia Jagdeo

fjagdeo@emeris.ac.za

P.A.S

Programme code	Mode of Delivery	Module Code	Module Name	Assessment Name	Assessment Date	Assessment Time	Semester	Applicable Prerequisite module	Submission Instruction
BCA1w (4 Year)	Distance	INSY6212	Information Systems 2B	Assignment 1	Monday, 07 September 2026	23h59	Semester 4		Submit on ARC
BCA1w (4 Year)	Distance	INSY6212	Information Systems 2B	Assignment 2	Monday, 05 October 2026	23h59	Semester 4		Submit on ARC
BCA1w (4 Year)	Distance	INSY6212	Information Systems 2B	ICEw	Lecturer set	Lecturer set	Semester 4		
BCA1w (4 Year)	Distance	INSY6212	Information Systems 2B	Written Unseen Exam	TBC	TBC	Semester 4		
BCA1w (4 Year)	Distance	INSY6212	Information Systems 2B	Written Unseen Supplementary Exam	TBC	TBC	Semester 4		

Integrated Curriculum Engagement (ICE)	
Minimum number of ICE activities to complete	4
Weighting towards the final module mark	10%

Assignments	Assignment 1	Assignment 2
Weighting	25%	30%
Duration	10 hours	10 hours
Total Marks	100	100
Period	3	4
Learning Units covered	LU1 – 4	LU1 – 6 (up to Theme 2)
Resources required	Refer to the assignment for information.	

Tests/Examination	Examination
Weighting	35%
Duration	2 hours
Total marks	100
Open/ closed book	Closed
Resources required	Calculator
Learning Units covered	ALL

Module resources

Textbook

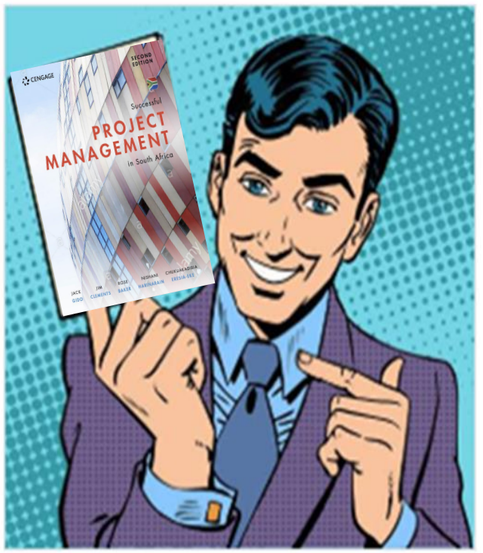
- Gido, J.; Clements, JP.; Baker, R.; Harinarain, N.; and Eresia-Eke, C. 2022. *Successful Project Management in South Africa.* 2nd edition. Cengage Learning. 9781473780415

Software and Additional Resources

- Microsoft Project, latest (e.g., 2016 Edition)
- The prescribed articles and resources for this module are in the MO

ARC

- Online content and activities
- Online Articles and Resources



Module outline

1. Introduction to Project Management
2. Project Identification and Selection
3. Project Proposal
4. Project Planning
5. Introduction to Project Software
6. Project Resource Utilisation and Schedule Development
7. Project Costs and Budgets
8. Project Risk Management
9. Project Manager, Project Teams and Organisational Structures

MODULE OUTCOMES:

MO1	Demonstrate Knowledge of the key concepts, principles and best practices of information technology project management in organization’s
MO2	Create an integrated project solution using effective information technology project management methodology
MO3	Demonstrate the use of project tools and techniques to support the management of project lifecycle an information technology project environment



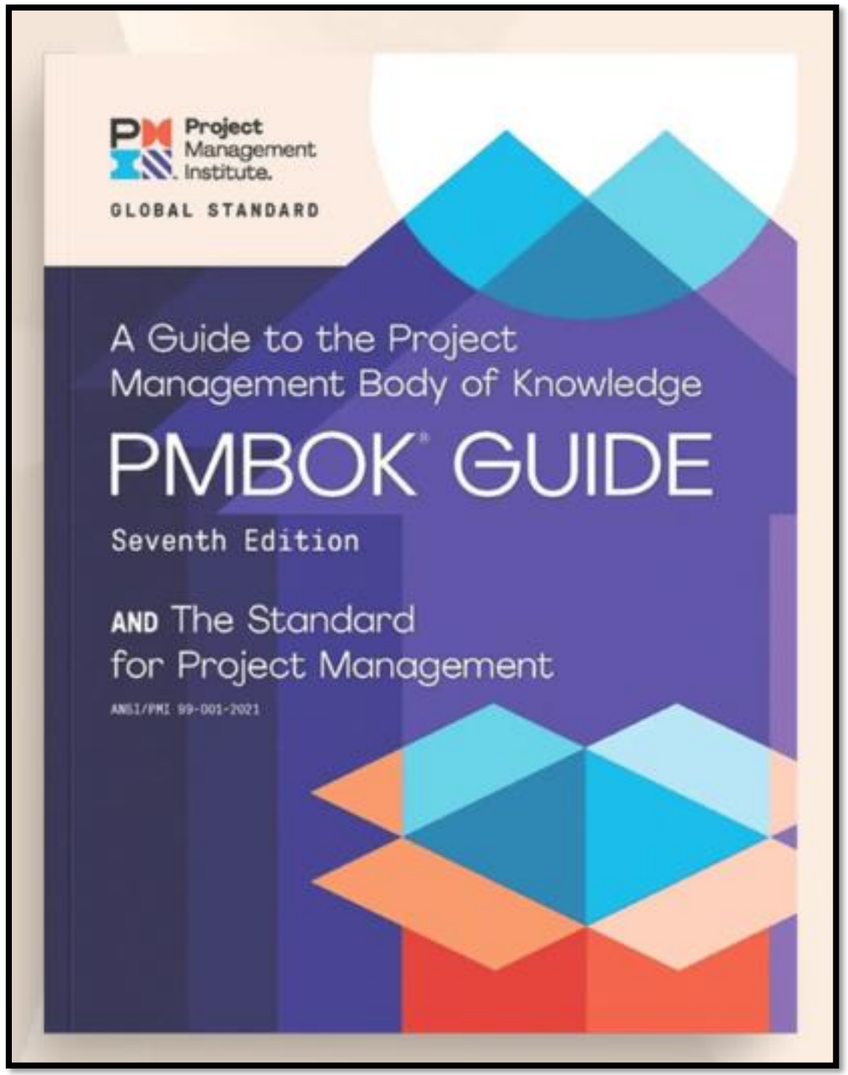
INTRODUCTION:



- Individuals need hard information and real skills to work successfully in a project environment and to accomplish project objectives.
- This module will challenge you to think critically about project management principles and to apply them within the context of the real world.
- All members/ people on the project team must have the knowledge and skills to work effectively together in a project environment.
- People do not become project managers by reading books; they become project managers by first being effective project team members.
- Therefore, this module/ course will provide the foundation that individuals need to be effective members of project teams and thereby boosting everyone's potential to rise to the challenge of managing teams and projects.

PROJECT MANAGEMENT BODY OF KNOWLEDGE

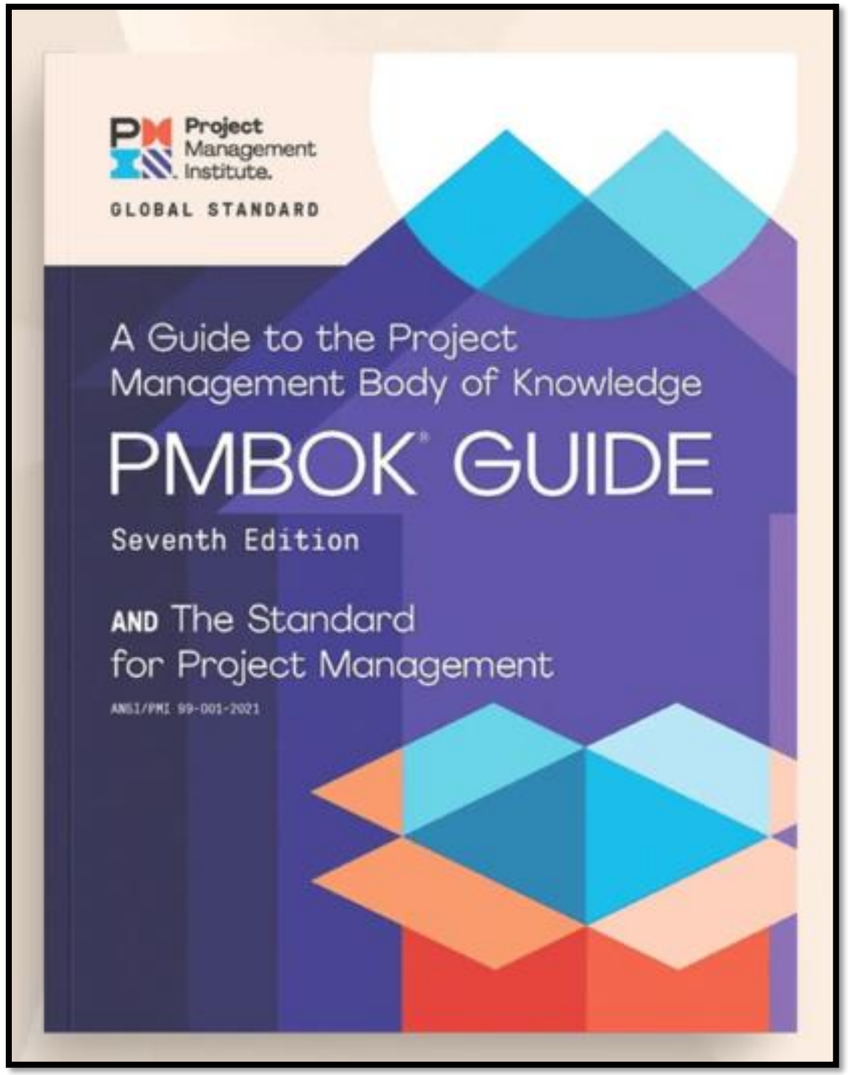
- What is the PMBOK Guide?
  - It's a **global standard** that provides best practices, guidelines, and tools to help manage projects effectively.



PROJECT MANAGEMENT BODY OF KNOWLEDGE

- Why is the PMBOK Guide Important?

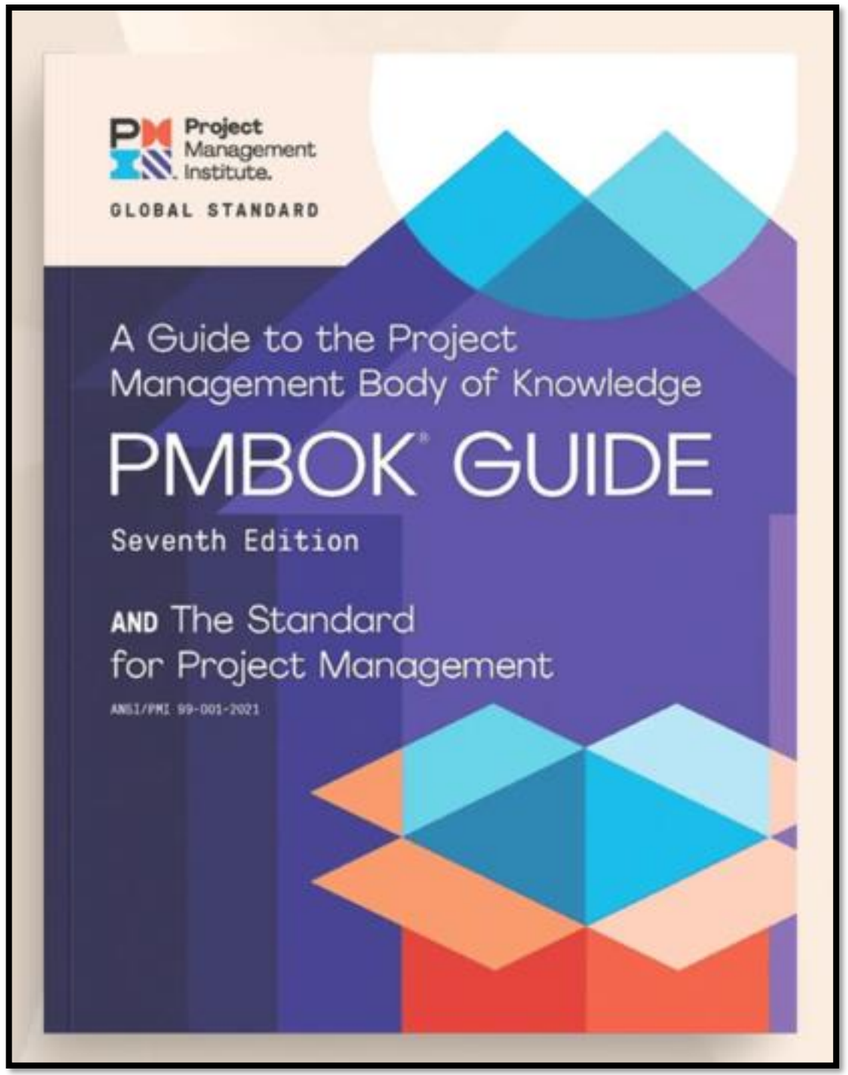
1. Standardisation
2. Reduces Project Failures
3. Widely recognised
4. Better resource management



PROJECT MANAGEMENT BODY OF KNOWLEDGE

- How is the PMBOK relevant to IT Project Management?

1. Project Initiation
2. Project Planning
3. Project Performing – Monitoring and Controlling
4. Project Closure



The 12 Principles of Project Management PMBOK 7th Edition

- Stewardship
- Team Leadership
- Stakeholder Engagement
- Value
- System Thinking
- Leadership



- Tailoring
- Quality
- Complexity
- Risk Management
- Adaptability and Resiliency
- Change

8 Project Performance Domains



Stakeholder



Team



Lifecycle



Planning



Project Work



Delivery

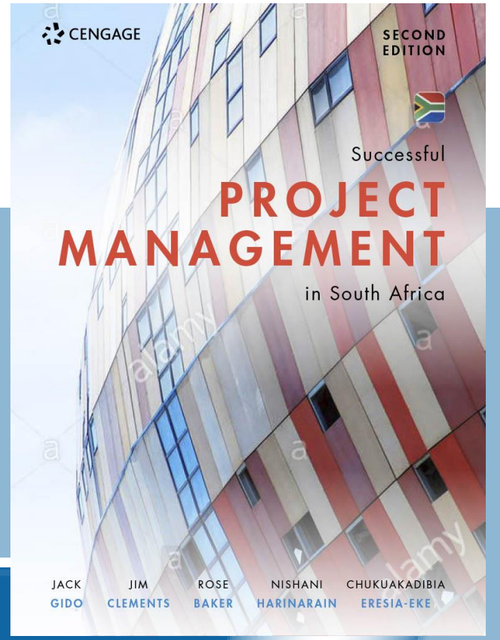


Measurement



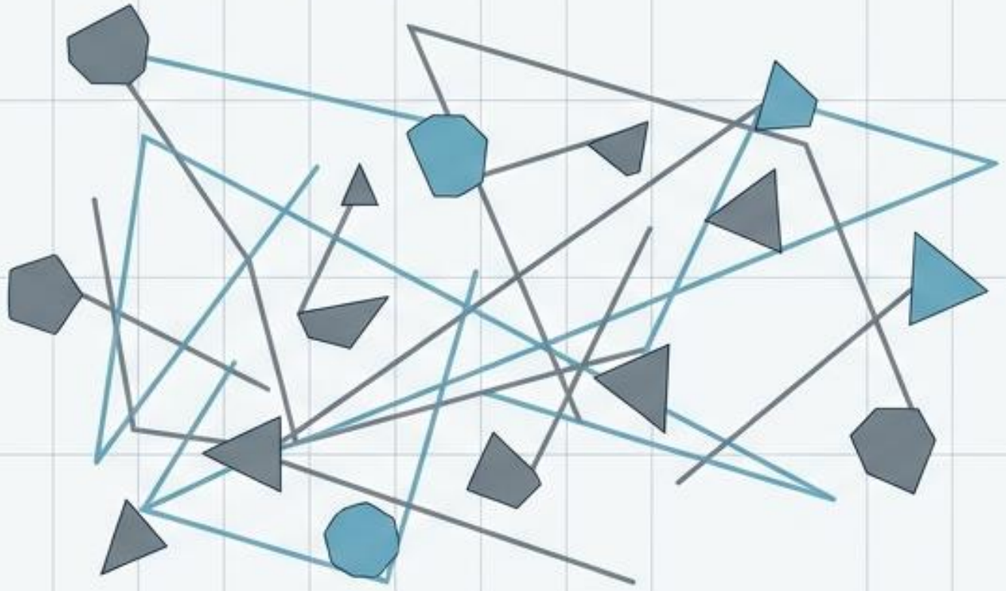
Uncertainty

CHAPTER 1



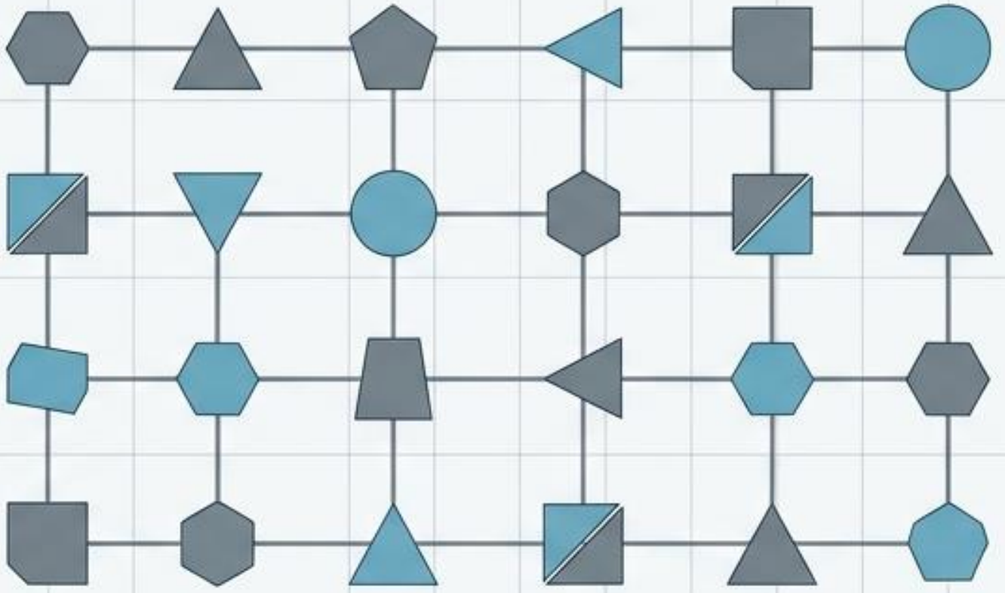
Project Management Concepts

Defining the Endeavour vs. The Discipline



The Project

An endeavour to accomplish a specific objective through a unique set of interrelated activities and the effective utilisation of resources.



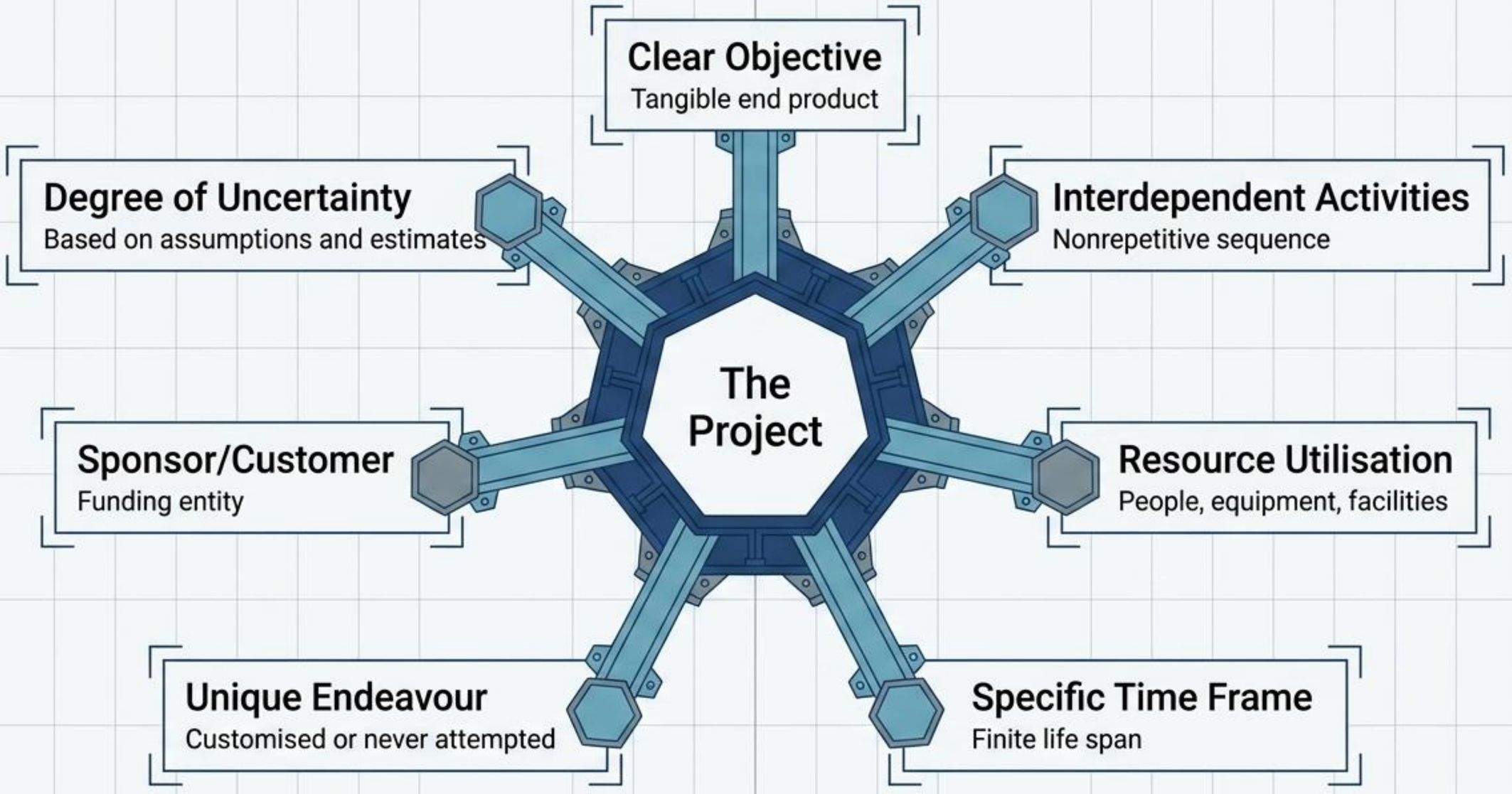
Project Management

The application of knowledge, skills, tools, and techniques to project activities in order to meet stakeholder needs and achieve the project objective.

The Objective: To deliver the tangible end product within schedule and budget.

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The Anatomy of a Unique Endeavour



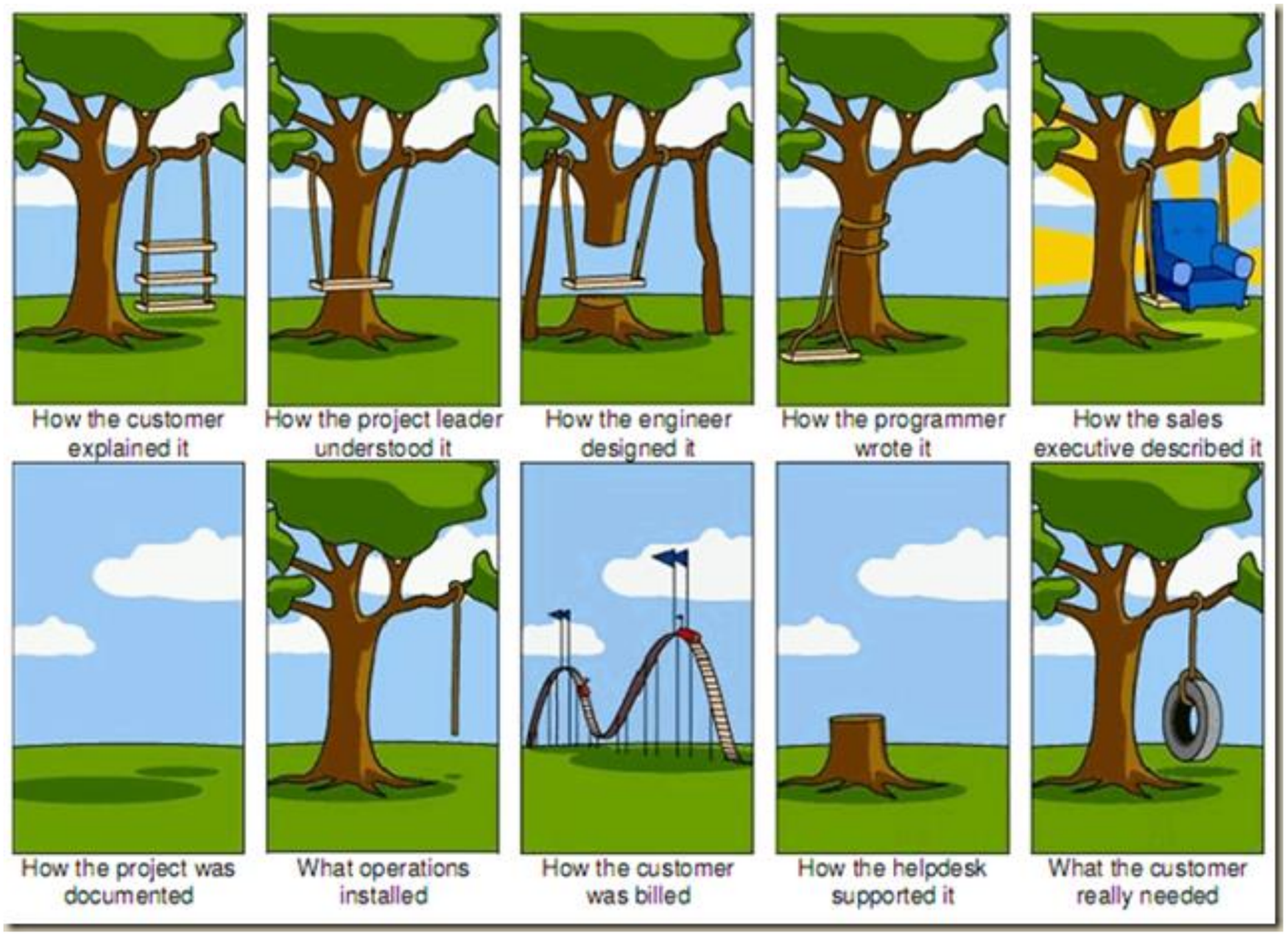
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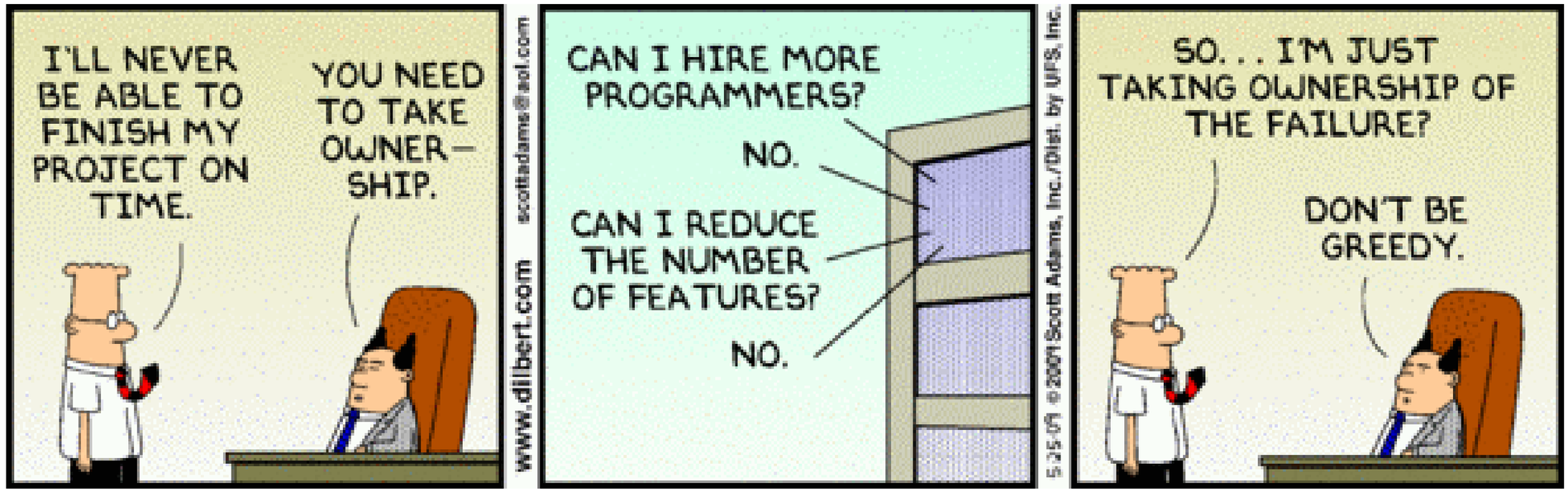
A project includes:

- Start and Finish
- Lifecycle
- Budget
- Non-repetitive
- Resources
- Single point of responsibility
- Fast-tracking
- Team roles

Project Management vs IT Project Management

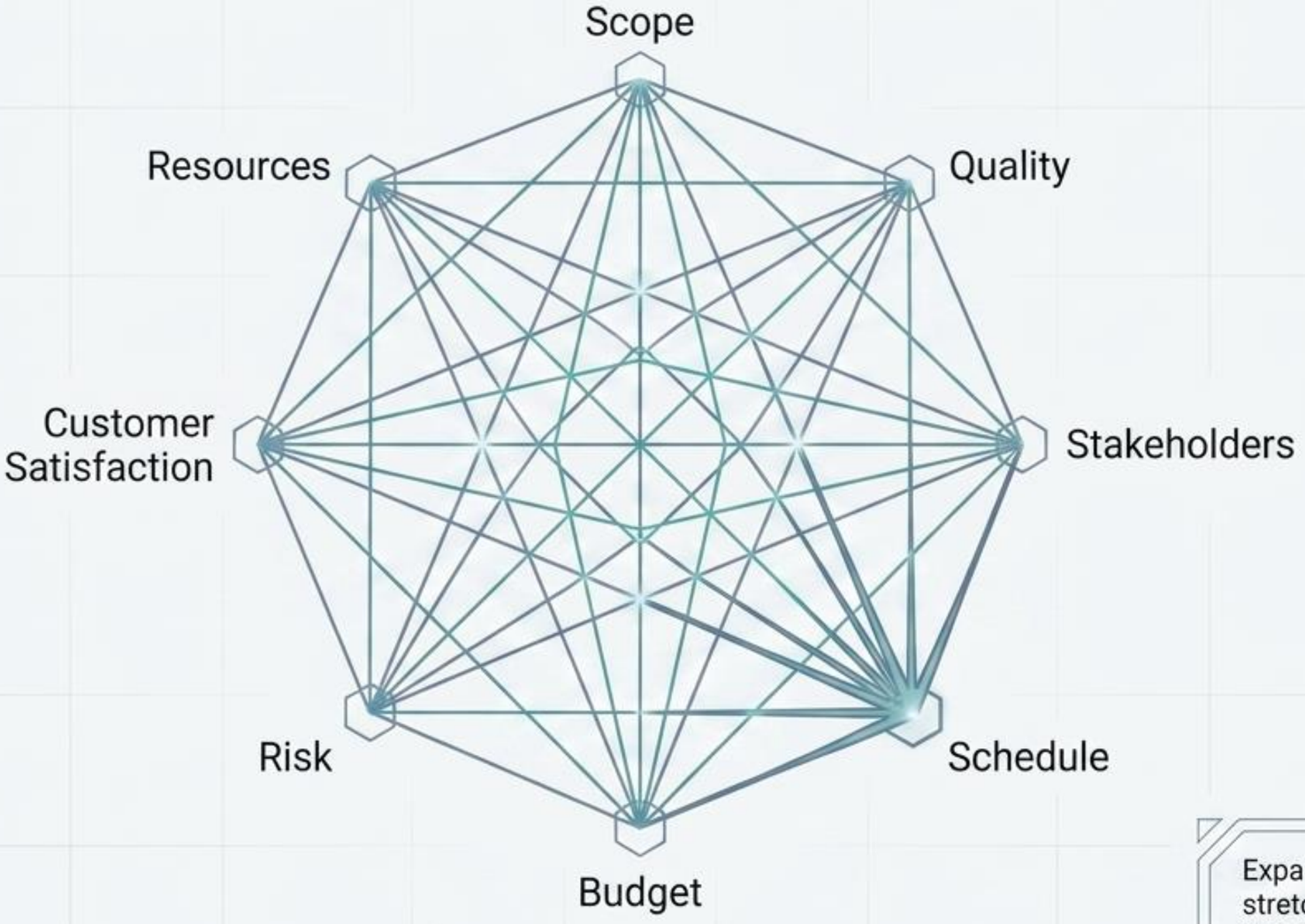
- Project management vs IT project management
- Is there a difference?
- If so, how are they different?





- Have you ever been in a situation where you could not complete a task because you were restricted or prevented to do so?
  - How would you deal with such a situation/s?

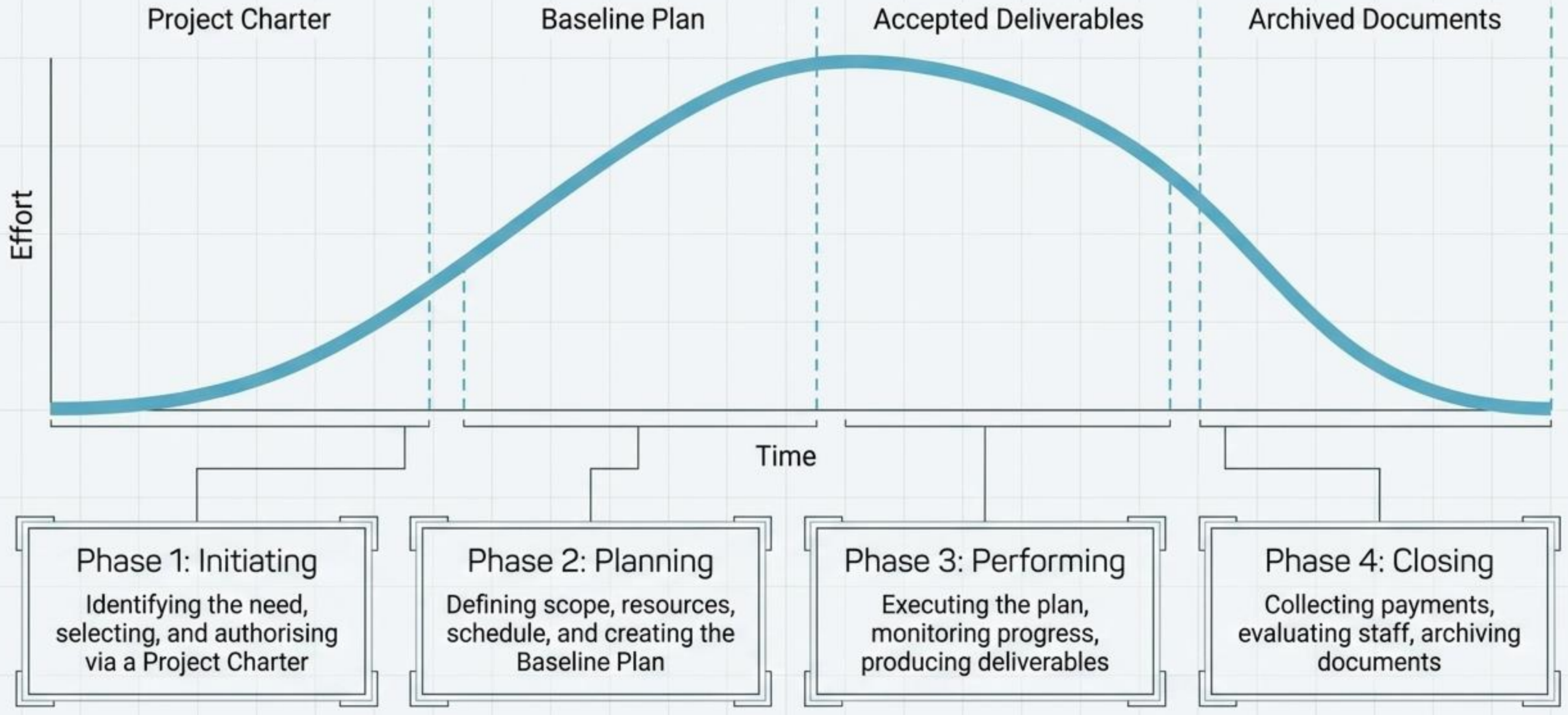
The Tension Web: Balancing Project Constraints



Expanding the scope inherently stretches the budget, time, or risk. Management is the act of maintaining equilibrium.

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The Effort Curve: Project Life Cycle



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Project Planning Process

- Establish project objective
- Define scope
- Create WBS
- Assign responsibility
- Define specific activities
- Sequence activities
- Estimate activity resources
- Estimate activity durations
- Develop project schedule
- Estimate activity costs
- Determine budget

FIGURE 1.3 Work Breakdown Structure

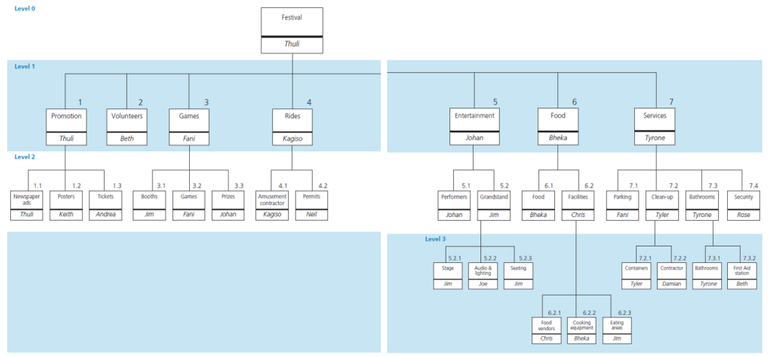


FIGURE 1.4 Network Diagram

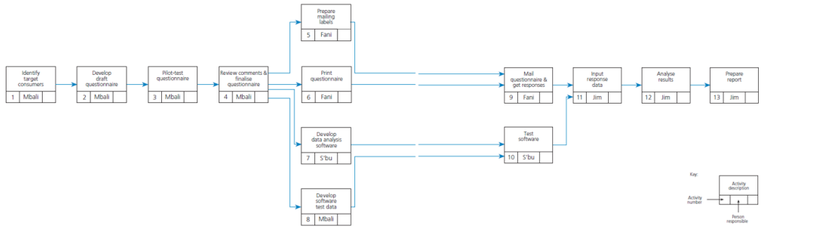
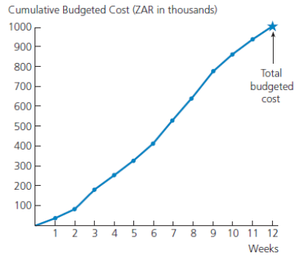


FIGURE 1.5 Project Schedule Consumer Market Study Project

Consumer market study project

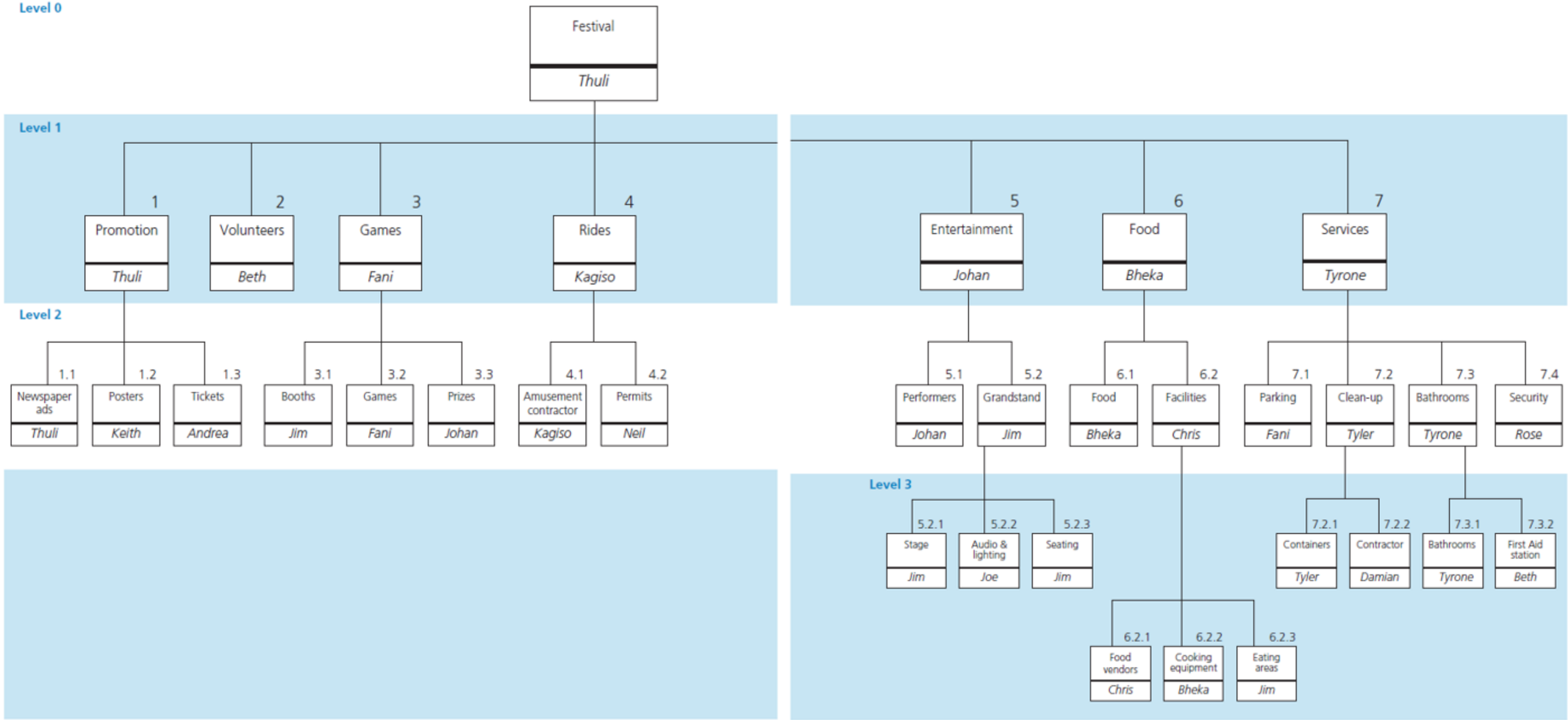
	Activity	Respon.	Dur. Estim.	Earliest Start	Earliest Finish	Latest Start	Latest Finish	Total Slack
1	Identify target consumers	Mbali	3	0	3	−8	−5	−8
2	Develop draft questionnaire	Mbali	10	3	13	−5	5	−8
3	Pilot-test questionnaire	Mbali	20	13	33	5	25	−8
4	Review comments & finalise questionnaire	Mbali	5	33	38	25	30	−8
5	Prepare mailing labels	Fani	2	38	40	38	40	0
6	Print questionnaire	Fani	10	38	48	30	40	−8
7	Develop data analysis software	S'bu	12	38	50	88	100	50
8	Develop software test data	Mbali	2	38	40	98	100	60
9	Mail questionnaire & get responses	Fani	65	48	113	40	105	−8
10	Test software	S'bu	5	50	55	100	105	50
11	Input response data	Jim	7	113	120	105	112	−8
12	Analyse results	Jim	8	120	128	112	120	−8
13	Prepare report	Jim	10	128	138	120	130	−8

FIGURE 1.6 Time-Phased Project Budget



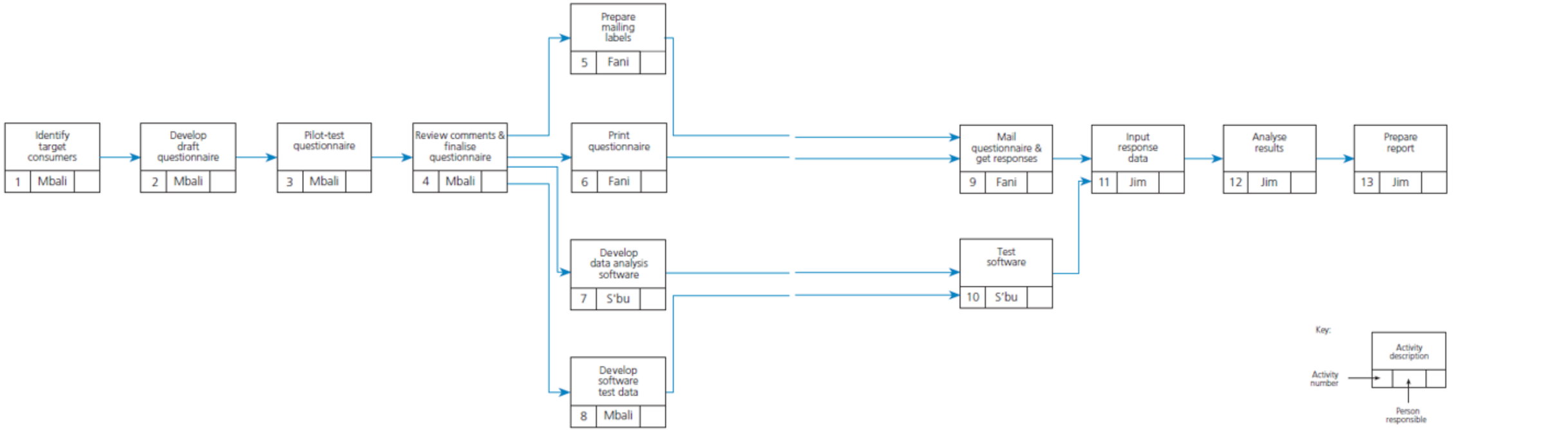
Create WBS and assign responsibility

FIGURE 1.3 Work Breakdown Structure



Sequence activities

FIGURE 1.4 Network Diagram



Develop the project schedule

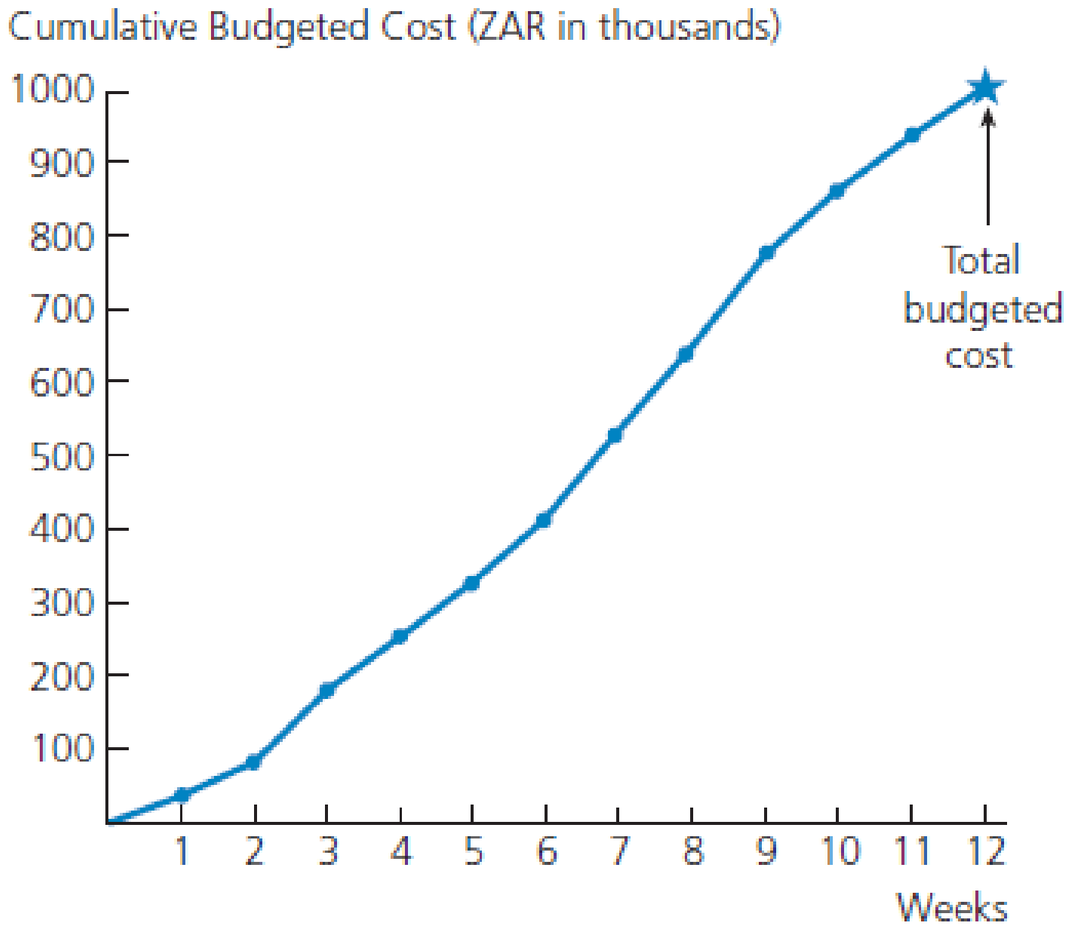
FIGURE 1.5 **Project Schedule Consumer Market Study Project**

Consumer market study project

	Activity	Respon.	Dur. Estim.	Earliest		Latest		Total Slack
				Start	Finish	Start	Finish	
1	Identify target consumers	Mbali	3	0	3	–8	–5	–8
2	Develop draft questionnaire	Mbali	10	3	13	–5	5	–8
3	Pilot-test questionnaire	Mbali	20	13	33	5	25	–8
4	Review comments & finalise questionnaire	Mbali	5	33	38	25	30	–8
5	Prepare mailing labels	Fani	2	38	40	38	40	0
6	Print questionnaire	Fani	10	38	48	30	40	–8
7	Develop data analysis software	S'bu	12	38	50	88	100	50
8	Develop software test data	Mbali	2	38	40	98	100	60
9	Mail questionnaire & get responses	Fani	65	48	113	40	105	–8
10	Test software	S'bu	5	50	55	100	105	50
11	Input response data	Jim	7	113	120	105	112	–8
12	Analyse results	Jim	8	120	128	112	120	–8
13	Prepare report	Jim	10	128	138	120	130	–8

Determine budget

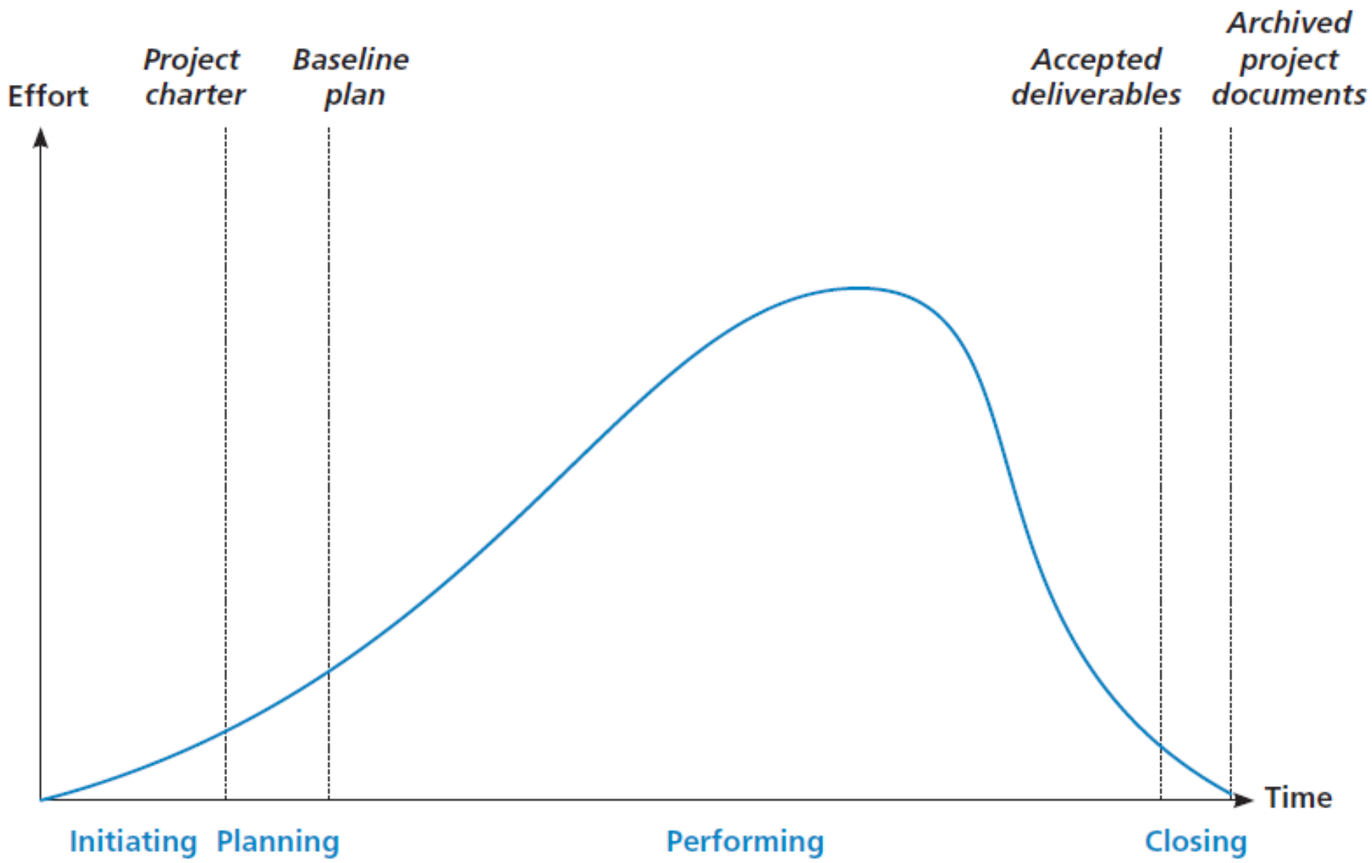
FIGURE 1.6 **Time-Phased Project Budget**



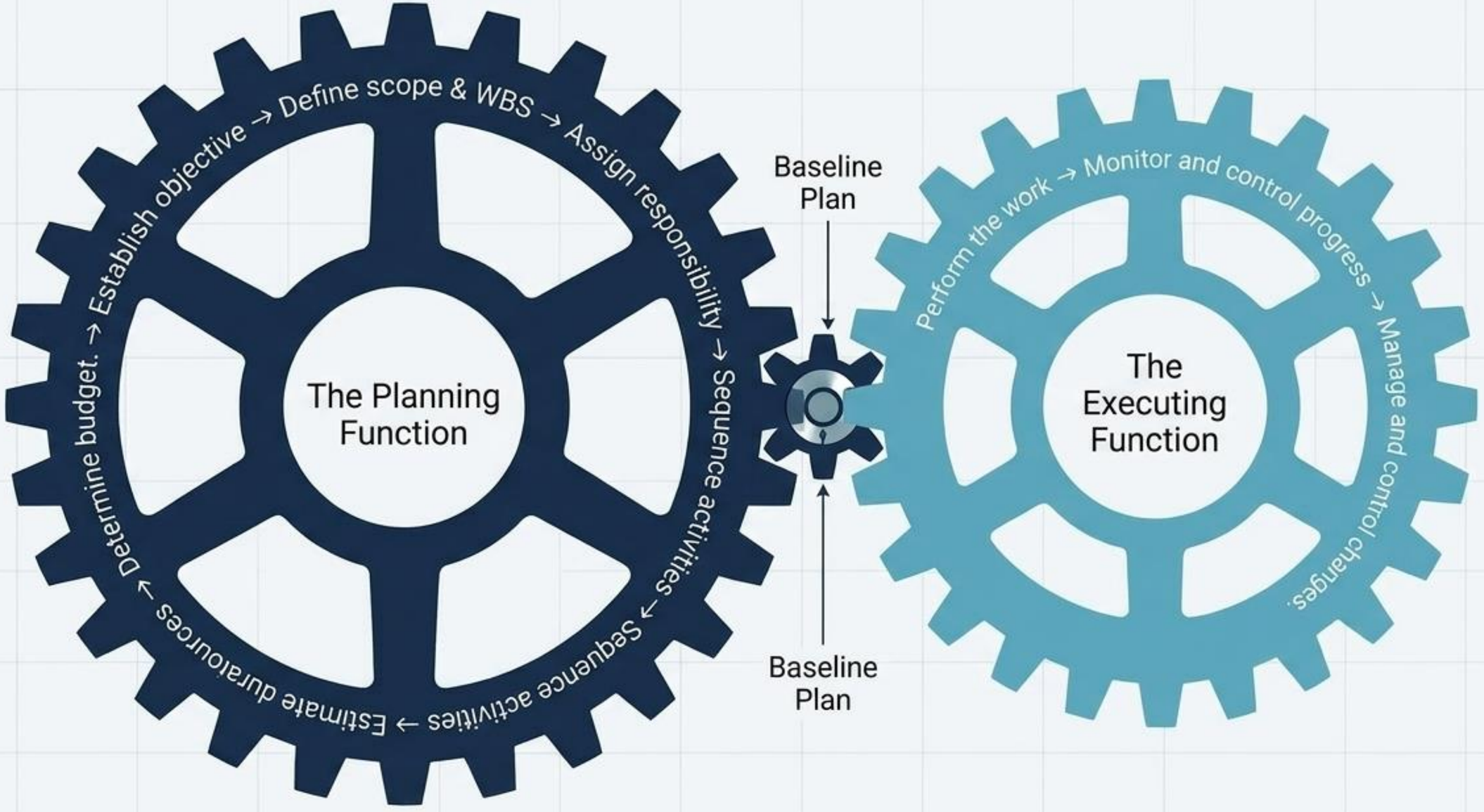
Execute the project plan

- Perform the work
- Monitor and control progress
- Control changes

FIGURE 1.2 Project Life Cycle Effort

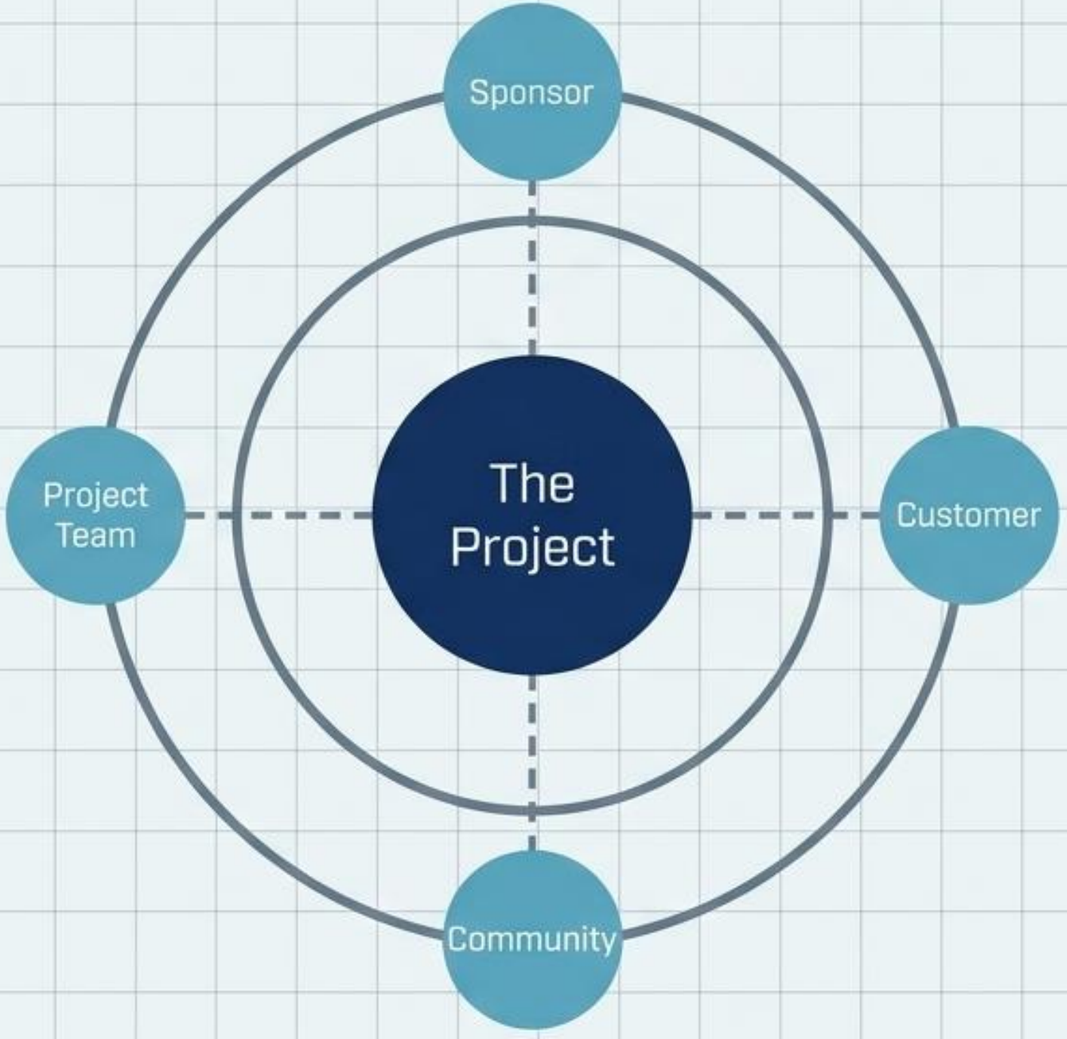


The Management Engine: Plan the Work, Work the Plan



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The Human Element: Stakeholders & Success Factors



Stakeholder Engagement

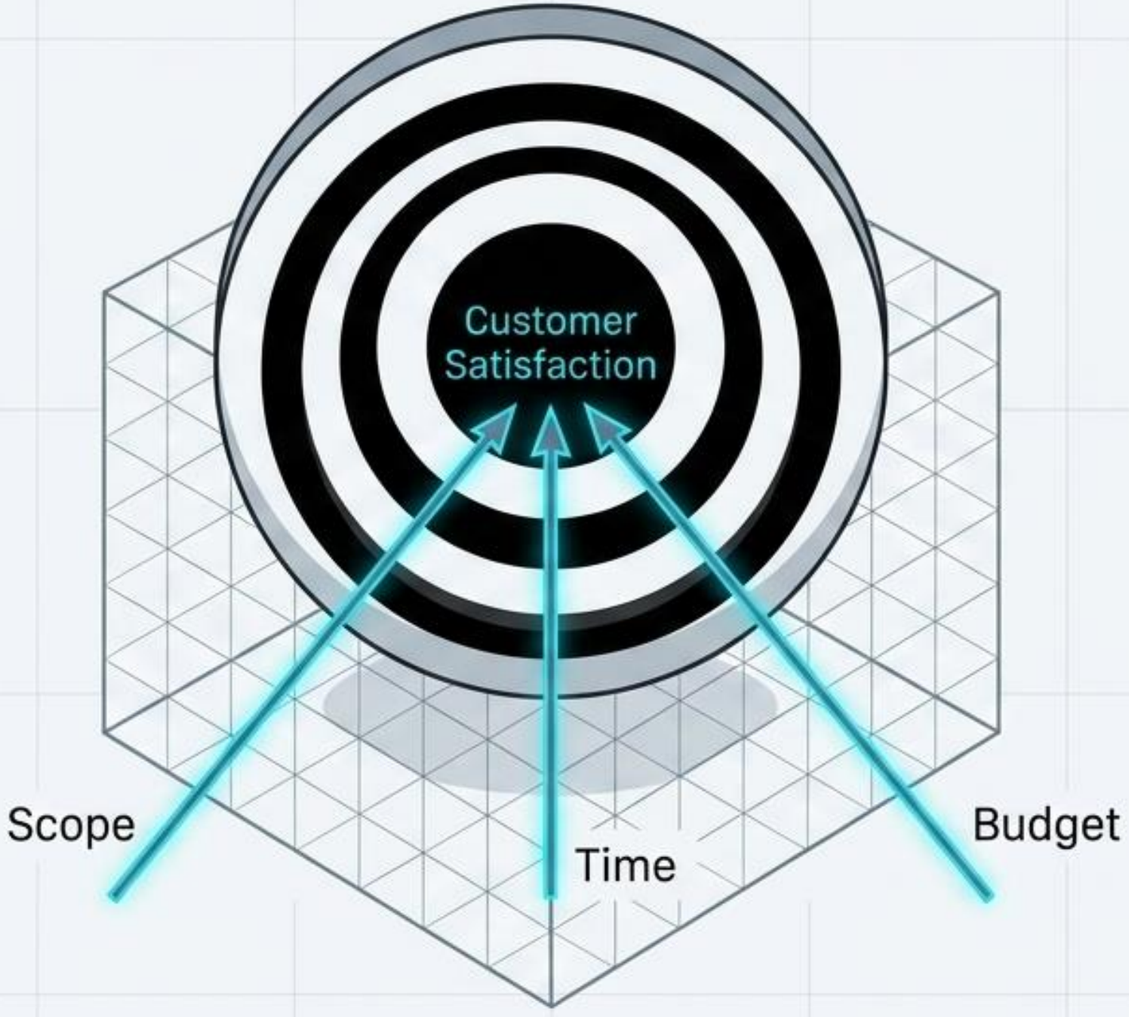
Building trust, understanding diverse cultures, and securing active participation from those influenced by the project.

Critical Success Factors:

1. Proactive planning and communication
2. Ongoing customer involvement
3. Timely measurement of actual vs. planned progress
4. Immediate corrective action

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The Ultimate Benefit of Project Management



Primary Outcome

A fully satisfied customer (or sponsor).

Value Delivered

Completing the full work scope in a quality manner, exactly on time, and strictly within budget.

Strategic Impact

The most efficient framework for introducing organisational change and delivering value to people.

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YOU'RE A WINNER, YOU'RE A WINNER ...
EVERYONE'S A WINNER!!!!!!
makeameme.org

LU1 - Theme 3: Project Management Guides, Frameworks and Methodologies

1. Differentiate between project management methodologies and standards
2. Explain the role and benefits of agile project management in an IT project
3. Discuss agile project management in the context of technology projects
4. How would you apply the different project management approaches and methodologies in managing an IT project

https://www.dummies.com/article/business-careers-money/business/project-management/agile-project-management-for-dummies-cheat-sheet-208332/

https://www.smartsheet.com/sites/default/files/Agile-PM-101-Beginners-Guide-Non-Project-Managers-Updated.pdf or https://www.smartsheet.com/blog/e-book-agile-project-management-101

http://www.sciencedirect.com/science/article/pii/S187704281402196X

The Matrix of Governance: Standards vs. Methodologies

Project Standards (The "What")	Project Methodologies (The "How")
Definition: A framework of established processes and guidelines (e.g., PMBOK).	**Definition:** A specific, actionable system of practices, techniques, and procedures (e.g., Agile, Waterfall).
Function: Defines the foundational knowledge, vocabulary, and best practices.	**Function:** Dictates the exact sequence, team structures, and execution philosophy.
Analogy: The dictionary and grammar rules of a language.	**Analogy:** The specific style or genre of a novel being written.

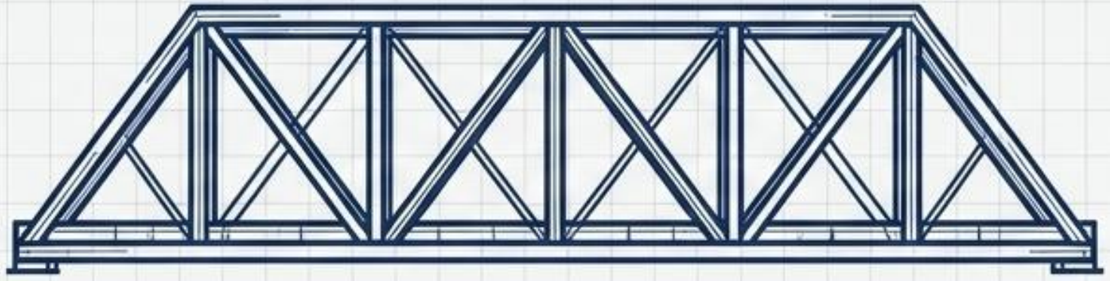
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Project Management Methodologies

Below are the most frequently used project management methodologies in the project management practice:

1. Adaptive Project Framework
2. Agile Software Development
3. Crystal Methods
4. Dynamic Systems Development Model (DSDM)
5. Extreme Programming (XP)
6. Feature Driven Development (FDD)
7. Information Technology Infrastructure Library (ITIL)
8. Joint Application Development (JAD)
9. Lean Development (LD)
10. PRINCE2
11. Rapid Application Development (RAD)
12. Rational Unified Process (RUP)
13. Scrum
14. Spiral
15. Systems Development Life Cycle (SDLC)
16. Waterfall (Traditional)

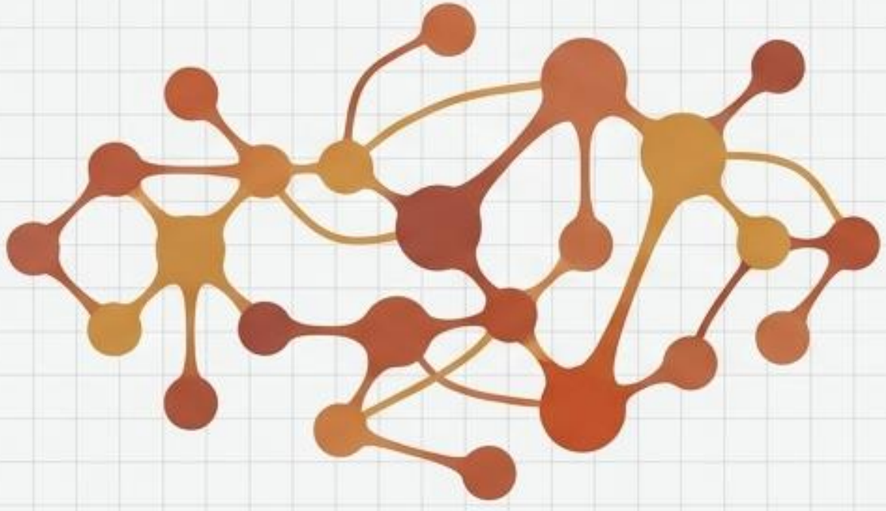
The IT Context: High Uncertainty & Rapid Change



Traditional Environment

Physical, predictable, and unchanging deliverables (e.g., construction).

Requirements are known upfront.



IT Environment

Deals with IT infrastructure, information systems, or software.

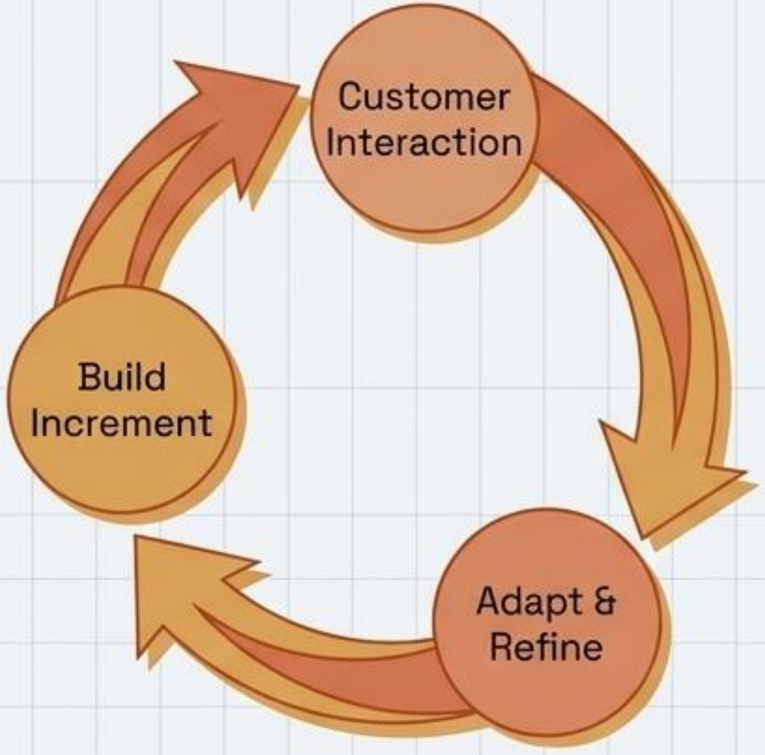
Unlike traditional projects, IT requirements are difficult to define upfront, highly volatile, and subject to rapid technological shifts.

Unforeseen circumstances (e.g., shifting user needs) can instantly jeopardise the objective if the methodology is too rigid.

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Agile: The Evolutionary Response to IT Complexity

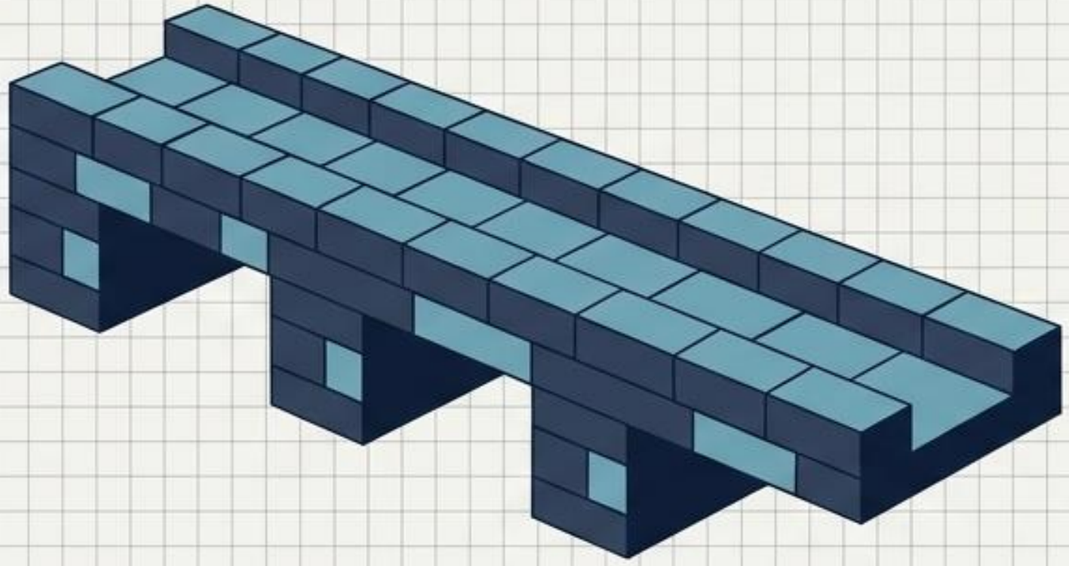
Definition: An approach designed to reduce development time and minimise risk in high-uncertainty environments.



Key Mechanics:
1. Continuous customer interaction
2. Small, self-organised teams
3. Short-time iterations (sprints)
4. Producing increments of working product features
5. Rapid adaptation to changing requirements

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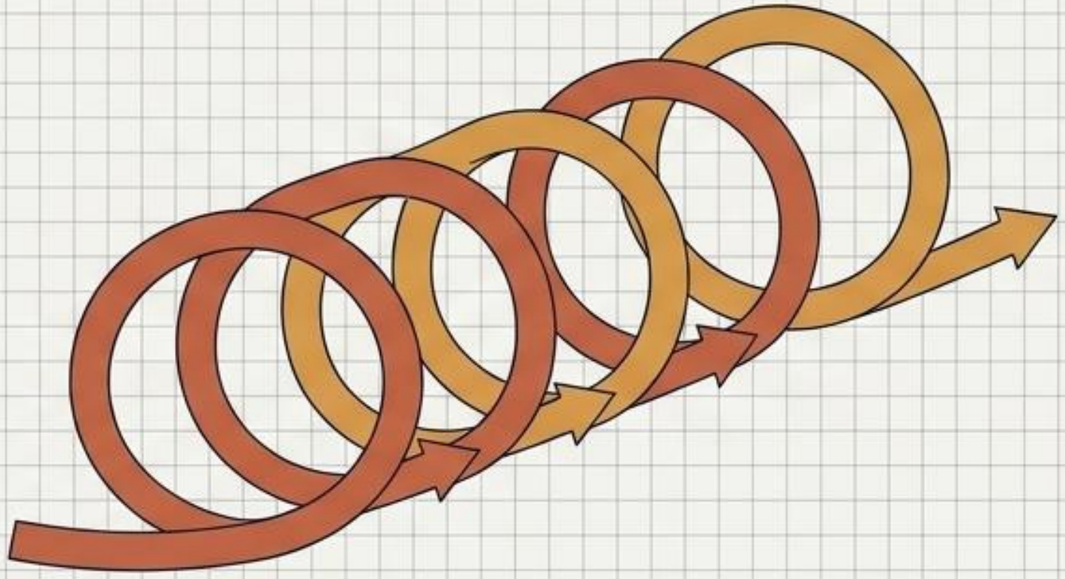
Applying the Playbook: Predictability vs. Adaptability



Traditional / Waterfall Application

Best for: Clear, unchanging physical deliverables (e.g., constructing a factory).

Execution: Sequential phases, heavy upfront planning, rigid change control.



Agile IT Application

Best for: Invisible, evolving deliverables (e.g., software development, digital platforms).

Execution: Incremental delivery, flexible scope, continuous integration.

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12 Agile Principles

The Agile Manifesto lists 12 principles to guide teams on how to execute with agility:

1 Satisfy the customer

Our highest priority is to satisfy the customer through early and continuous delivery of valuable software.

2 Welcome changing requirements

Even late in development. Agile processes harness change for the customer's competitive advantage.

3 Deliver working software frequently

from a couple of weeks to a couple of months, with preference to the shorter timescale.

4 Work together

Business people and developers must work together daily throughout the project.

5 Build projects around motivated individuals

Give them the environment and support they need, and trust them to get the job done.

6 Face-to-face conversation

The most efficient and effective method of conveying information to and within a development team is face-to-face conversation.

7 Focus on working software

Working software is the primary measure of progress.

8 Promote sustainable development

Agile processes promote sustainable development. The sponsors, developers, and users should be able to maintain a constant pace indefinitely.

9 Ensure technical excellence

Continuous attention to technical excellence and good design enhances agility.

10 Simplicity

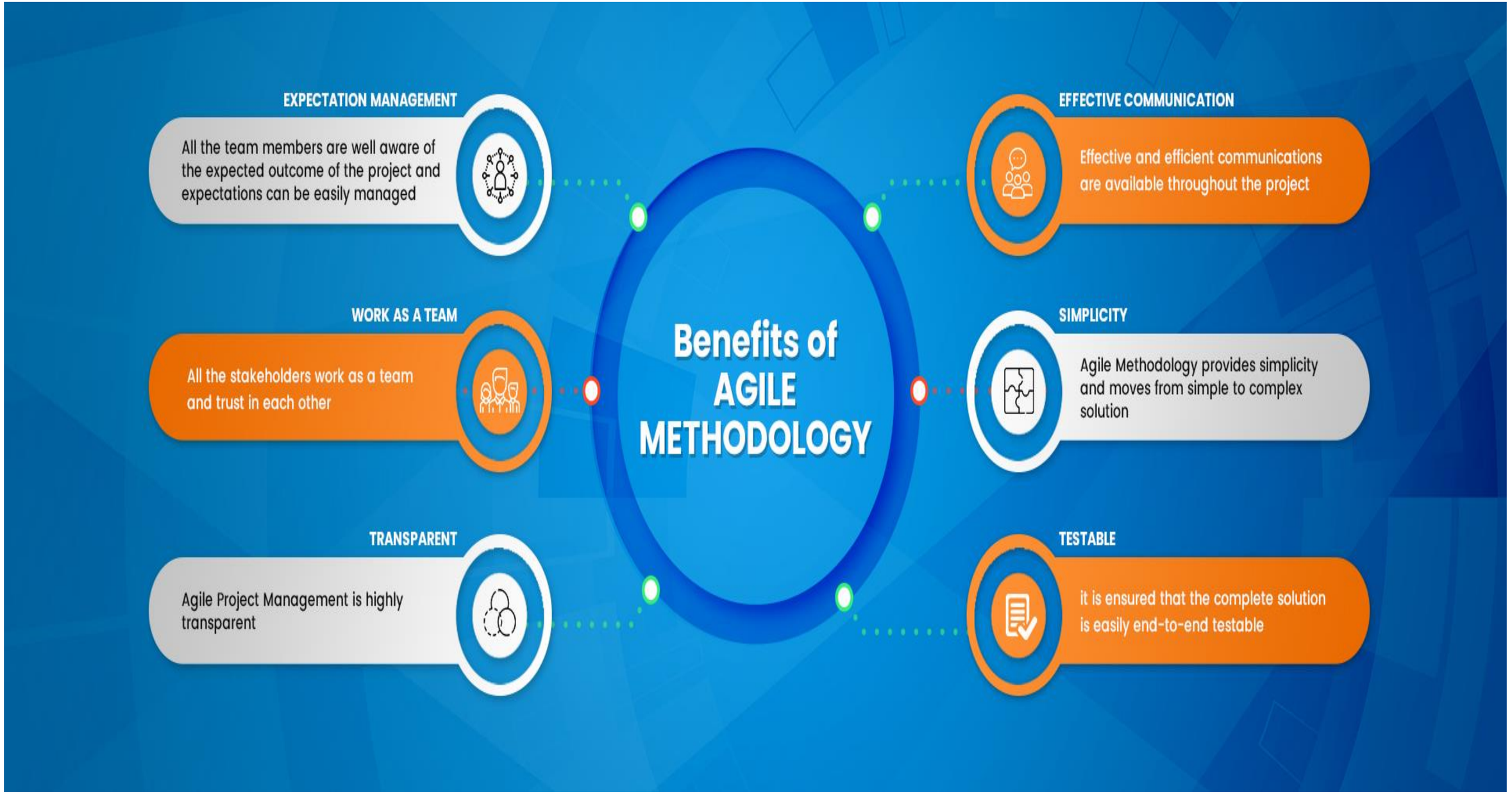
The art of maximizing the amount of work not done – is essential.

11 Self-organizing teams

The best architectures, requirements, and designs emerge from self-organizing teams.

12 Reflect and adjustment

At regular intervals, the team reflects on how to become more effective, then tunes and adjusts its behavior accordingly.


Benefits of AGILE METHODOLOGY
EXPECTATION MANAGEMENT
All the team members are well aware of the expected outcome of the project and expectations can be easily managed
WORK AS A TEAM
All the stakeholders work as a team and trust in each other
TRANSPARENT
Agile Project Management is highly transparent
EFFECTIVE COMMUNICATION
Effective and efficient communications are available throughout the project
SIMPLICITY
Agile Methodology provides simplicity and moves from simple to complex solution
TESTABLE
it is ensured that the complete solution is easily end-to-end testable

Top Methodologies Used to Implement Agile

There are a number of specific methods within the Agile movement. We'll take an in-depth look at two of the top Agile Methodologies: Scrum and Kanban.



Scrum



Kanban

Scrum Methodology

Scrum is a subset of Agile and one of the most popular process frameworks for implementing Agile.

It is an iterative development model often used to manage complex software and product development. Fixed-length iterations, called sprints lasting one to two weeks long, allow the team to ship software on a regular cadence. At the end of each sprint, stakeholders and team members meet to plan next steps.



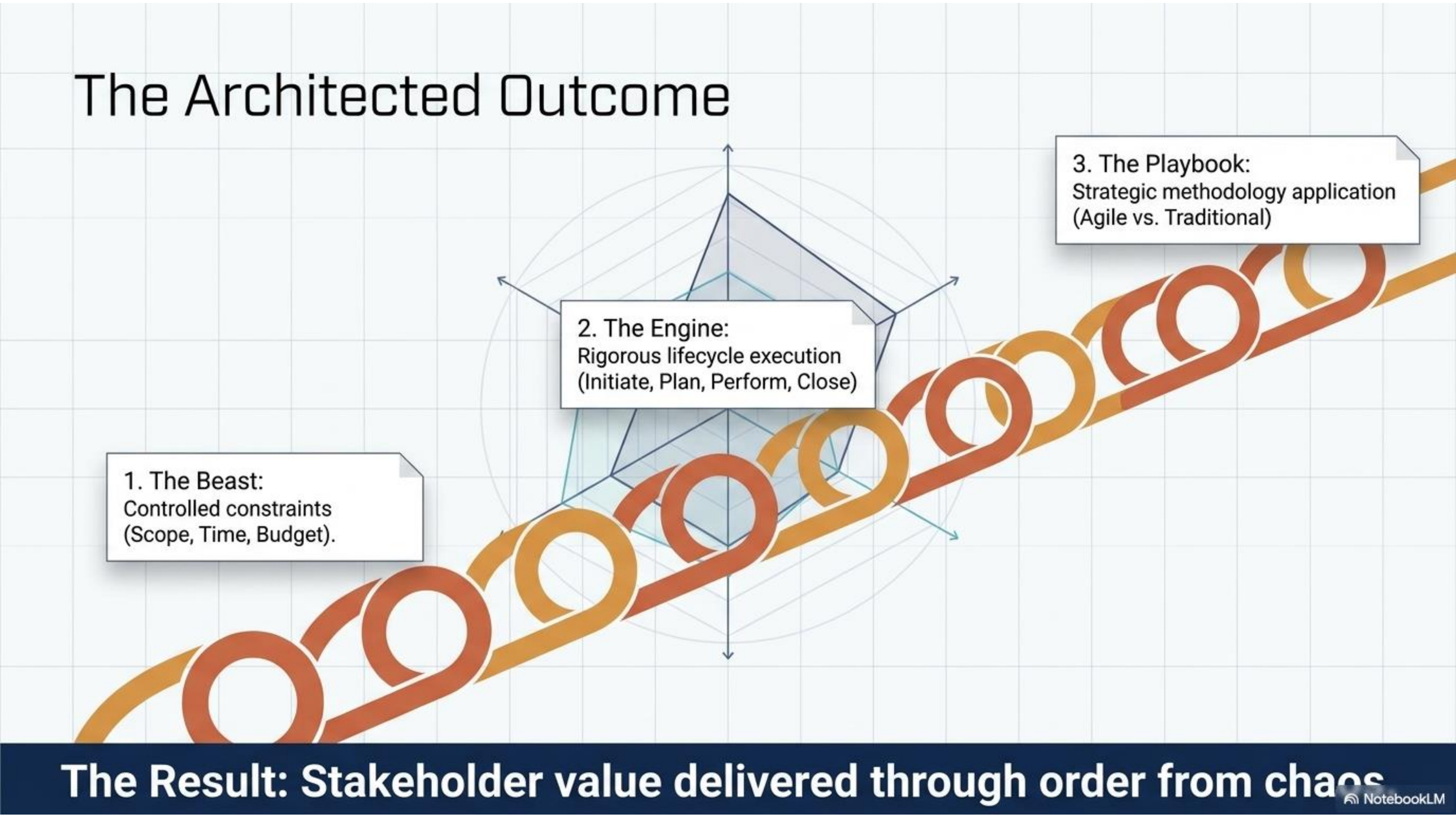
Kanban Methodology

Kanban is Japanese for "visual sign" or "card." It is a visual framework used to implement Agile and shows what to produce, when to produce it, and how much to produce. It encourages small, incremental changes to your current system and does not require a certain set up or procedure (meaning, you could overlay Kanban on top of other existing workflows).

Kanban was inspired by the Toyota Production System and Lean Manufacturing. In the 1940s, Toyota improved its engineering process by modeling it after how supermarkets stock shelves. Engineer Taiichi Ohno noticed that supermarkets stock just enough product to meet demand. Inventory would only be restocked when there was empty space on the shelf (a visual cue).

These same ideas apply to software teams and IT projects today. In this context, development work-in-progress (WIP) takes the place of inventory, and new work can only be added when there is an "empty space" on the team's visual Kanban board. Kanban matches the amount of WIP to the team's capacity, improving flexibility, transparency, and output.

The Architected Outcome



The Result: Stakeholder value delivered through order from chaos

NotebookLM